Credit Risk Portfolio Analytics | Solution Overview

Executive Brief: Dashboard Architecture, AI Governance, Default Modeling, Insights & Production Roadmap

1. Dashboard Architecture & View-by-View Analyst Workflow

• **Portfolio Overview:** Core portfolio metrics (14,685 accounts, 16.10% default rate, $512.6M exposure), product delinquency bars, predicted PD risk bands (<10%, 10–25%, ≥25%), and monthly origination vintage curves.
• **Segmentation & Insights:** Profile distributions across Income, Employment, Region, Age, Loan Type; plus interactive **Employment Type x Income Band** cross-tab heatmap.
• **Account Risk & SHAP:** Searchable borrower dossier with predicted PD, terms, repayment history, and **local feature contributions (SHAP proxy)** explaining exact risk drivers.
• **Ask the Portfolio (LLM):** Conversational interface answering slice queries without hallucinated figures.
• **Anomaly Review Queue:** Case investigation queue with transparent trigger reasons and persistent decision recording.
• **Model Monitoring:** Holdout discrimination curves, decile calibration (E/A), and post-deployment stability protocols.

2. Mandatory Embedded AI Features Governance Specification

AI Feature	Core Mechanism	Operational Trigger	False Positive (Risk)	False Negative (Risk)	Human-in-the-Loop Step
Ask the Portfolio (LLM)	Groq LLM (openai/gpt-oss-20b) queried over slice JSON aggregate metrics. Strict anti-hallucination prompt.	Analyst types question and clicks 'Submit Query' in Tab 4.	Model misinterprets slice denominator or asserts speculative causation.	Model excessively declines a valid question inferable from slice data.	Analyst verifies reported figures against displayed dashboard KPI cards before policy decisions.
Anomaly Detection	Multivariate IsolationForest + peer group baselines (Loan Type x Income) + DPD trajectory heuristics.	Automated background calculation on filtered cohort.	Prime borrower with irregular seasonal cash flow flagged as credit distress.	Gradual delinquency deteriorating just below contamination cut-off.	Analyst inspects reason flags and logs **Confirm / Dismiss / Escalate** decision to audit log.

3. Default Prediction Modeling Results & Post-Deployment Monitoring

• **Methodology:** 70/30 stratified holdout (10,279 train / 4,406 test). Evaluated Logistic Regression baseline against Random Forest & LightGBM using 59 features spanning bureau history, 12-month repayment trajectory, collections activity, and borrower leverage.
• **Discrimination Results: Random Forest achieved Test AUC: 0.6609, KS: 0.2342, Gini: 0.3217** (Baseline Logistic Regression achieved Test AUC: 0.6565, KS: 0.2308, Gini: 0.3131).
• **Key Drivers:** Top features by Gini importance and SHAP are annual_income (0.064), repayment_mean_dpd (0.047), years_employed (0.045), total_debt_to_income (0.043), and bureau_mean_dpd (0.040).
• **Monitoring Plan:** Monthly Population Stability Index (PSI) tracking (baseline-to-test PSI: 0.0250, well within <0.10 Green status). Automated alerts trigger if rolling 3-month AUC drops below 0.65, KS drops below 0.20, or score decile E/A calibration deviates beyond ±20%.

4. Top 3 Empirical Portfolio Insights & Strategic 'So What' Directives

1. Steep Income Inversion Gradient (4.0x Risk Multiplier): Borrowers earning <$30k default at **38.50%** (174/452) vs **9.53%** (293/3,076) for $100k+ earners. *So What:* Mandate max 35% DTI caps for sub-$30k applicants, require alternate income verification, and reduce unsecured credit limits.
2. Extreme Capital Exposure in Mortgages (24.65% Default Rate): Mortgages demonstrate the highest product default rate (353/1,432) while carrying an average balance of $219,605 ($314.5M total original balance). *So What:* Conduct immediate collateral Loan-to-Value (LTV) audits, increase property appraisal haircuts, and tighten broker origination standards.
3. Bivariate Compounding Risk (Unemployed x Low Income Spike): In the cross-dimensional intersection of Unemployed borrowers earning <$30k, default reaches **44.10%**, compared to **8.99%** for Salaried $100k+ borrowers. *So What:* Enforce bivariate underwriting stop-screens automatically rejecting unsecured credit for <$30k non-salaried applicants.

5. Core Assumptions & Bank IT/Data Team Production Requirements

• **Assumptions:** Month-12 closing balance represents latest exposure; observation window is a standardized 12-month post-origination performance period; null bureau closed dates denote active open facilities; synthetic data distributions reflect underlying macroeconomic stability.
• **IT & Data Requirements:** 1) *Point-in-Time Feature Store* (e.g. Feast/Hopsworks) for leak-free historical snapshots; 2) *Enterprise Secrets Vault* (e.g. AWS Secrets Manager / Azure Key Vault) for LLM API credentials; 3) *Durable Audit Database* (PostgreSQL) for regulatory storage of analyst review decisions; 4) *Model Risk Governance (SR 11-7)* approval and automated monthly PSI/CSI pipeline execution.